

MFTrans: A Multi-Resolution Fusion Transformer for Robust Tumor Segmentation in Whole Slide Images

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Abstract

Accurate tumor segmentation in whole slide image (WSI) is essential for histopathological diagnosis and research, but the traditional manual analysis is labor-intensive and prone to variability. Furthermore, many artificial models focus on specific magnification images, limiting the detailed information available for segmentation. To address these challenges, we propose MFTrans, a novel multi-resolution fusion transformer with a CNN-based architecture designed for efficient tumor segmentation in WSI. Inspired by the diagnostic procedures of expert pathologists, MFTrans integrates both high- and low-magnification images, capturing detailed local features and broader contextual relationships through a dual-branch architecture. The model employs a global token transformer and cross-attention mechanism to fuse hierarchical features from dual branches to improve segmentation performance. We evaluate MFTrans on three real-world WSI datasets: Camelyon16, PAIP2019, and Catholic Uijeongbu St. Mary's hospital dataset, demonstrating its superior segmentation performance over state-of-the-art methods in balanced and imbalanced setups. These results highlight MFTrans's effectiveness in medical image analysis and its generalizability across different datasets, making it a robust tool for automated cancer diagnostics. Our code is available at <https://github.com/aimed-gist/MFTrans>.

1. Introduction

Histopathology, the examination of tissue samples to study disease, is crucial in clinical practice for accurate diagnosis. Pathologists manually analyze histopathological samples, establishing the gold standard for cancer diagnostics and prognosis [1, 15, 23]. However, manual analysis is time-consuming and subject to variability due to human factors, with studies indicating up to a 24% discrepancy in

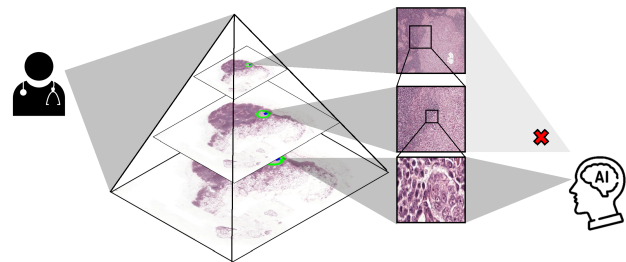


Figure 1. **Motivations.** Expert pathologists assess tumor structures at lower magnifications and examine individual cell morphology at higher magnifications. However, current models often focus on specific magnifications, which limits the detailed information used for segmentation.

interpretations among experts [13].

Recent advancements in artificial intelligence (AI) technology have led to success in various applications, particularly in natural language processing and computer vision [10, 29, 31]. In computer vision, Vision Transformer (ViT) [12] and its extensions [5, 16, 27] have demonstrated state-of-the-art performance on various benchmarks [8, 26, 40, 42]. Specifically, ViT-based models with the self-attention module provide more effective global contextual representations by modeling long-range relationships through pairwise interactions between token embeddings [12]. In medical image analysis, U-Net is the first segmentation method for biomedical images that utilized a CNN-based encoder and decoder [32], and UnetR employed ViT as its encoder to extract feature [17]. Specifically, CNNs are highly efficient at handling local features due to their convolutional nature, which excels at extracting fine-grained features at a regional level. Other approaches [7, 16, 41] have attempted to leverage the power of CNN and ViT to improve segmentation performance. In analyzing histopathology, the collected tissue sample undergoes several steps, including fixation, slicing, and staining, to create pathology glass slides. Pathol-

ogists then examine cells under microscopy at high magnification, known as whole slide image (WSI). A significant challenge of WSI analysis is high computational consumption due to its unusually high resolution [19]. Expert pathologists typically assess tumor architecture at lower magnifications and observe the morphology of individual tumor cells at higher magnifications, such as nuclear to cytoplasmic ratio and nuclear features (Fig. 1) [22]. However, most AI studies typically focus on patches of specific magnification or resolutions (e.g., low-resolution or middle-resolution images) that contain far fewer details.

In this study, we propose a multi-resolution fusion transformer (MFTrans) that utilizes both high- and low-magnification WSI to segment tumors. This approach is inspired by the clinical behavior of pathologists, who alternate between different magnifications during diagnosis. Our scientific contributions are summarized as follows:

- We extend a cross-attention mechanism to integrate complementary multi-resolution WSI. We aimed to learn shared and distinct representations across multi-resolution WSI information.
- We develop an efficient Global Token Transformer to enhance the transformer’s ability to capture and integrate multi-resolution information from WSI more effectively.
- We perform rich experiments on three real WSI datasets to demonstrate the effectiveness and clinical benefits of the proposed model over the competing algorithms.

The rest of this paper is organized as follows. In Sec. 2, we describe the state-of-the-art AI methods. In Sec. 3, we describe our proposed model (i.e., MFTrans). In Sec. 4, we present the experimental setup and results on three real WSI datasets and provide a brief discussion about experimental results. In Sec. 5, we summarize and describe the potential implications of the study.

2. Related Work

CNN-based methods For the field of medical image segmentation, CNNs with a U-shaped structure, commonly known as U-Net [32], and its extensions showed superior performance [2, 25, 37, 39]. This success was attributed to the skip connections that allowed the network to combine low-level spatial information with high-level contextual information, thereby enhancing segmentation accuracy and performance. Zhou *et al.* Proposed U-Net++ utilized nested and dense skip pathways to enhance feature propagation further and reduce the semantic gap between the encoder and decoder [43]. Attention U-Net integrated attention mechanisms into the U-Net architecture to focus on the

most relevant features [29]. Recently, U-Net 3+ was proposed to incorporate full-scale skip connections and multi-scale feature aggregation [18]. This model significantly improved segmentation accuracy while maintaining computational efficiency.

Transformer-based methods Based on the success of ViTs architecture, researchers applied it to medical image segmentation [9, 11, 14, 20, 24, 30]. Compared with CNN-based models, ViT models perceived the entire image as a sequence of patches, making them adept at understanding overall tissue morphology and the spatial distribution of tumor cells across the slide. TransUNet utilized a ViT encoder within a U-shaped structure [7], while ViT-V-Net extended this approach to volumetric medical data [6]. The Medical Transformer (MedT) employed a gated axial-attention mechanism to extract high-level global information and fine-grained local details separately [36]. SwinUNet and Swin-UNETR learned hierarchical feature representation through shifted windows with local attention [4, 16]. These methods demonstrated the potential of ViTs in capturing complex patterns and enhancing segmentation accuracy in medical imaging. However, fully ViT-based models require high computational costs and large labeled datasets, which are not suitable for the medical domain.

Dual-branch methods To overcome high computational costs and the need for large labeled datasets, researchers proposed dual-branched approaches integrating CNNs and transformers. These methods aimed to combine the local feature extraction capabilities of CNNs with the global context modeling of transformers, enhancing the overall performance of medical image segmentation. TransFuse is an early dual-branch model combining CNNs for local feature extraction and transformers for capturing global context and fusing CNN and transformer features with various pooling attention [41]. DHUNet also utilized a dual branch approach with CNNs and transformers to extract features separately [38]. However, it merged feature representation via transposed convolution feature concatenation at multiple levels to enhance segmentation accuracy. Although these approaches showed improved segmentation performance, existing methods only focused on a specific resolution of WSI that contained far fewer details.

3. Methodology

3.1. Overall architecture

The proposed method (MFTrans, a multi-resolution fusion transformer) consists of three components: a global branch, a local branch, and a fusion branch, as shown in Fig. 2. The global branch extracts global relationships between patches in low magnification WSI, passing through stages composed of token and patch merging layers and transformer blocks. The local branch provides regional rep-

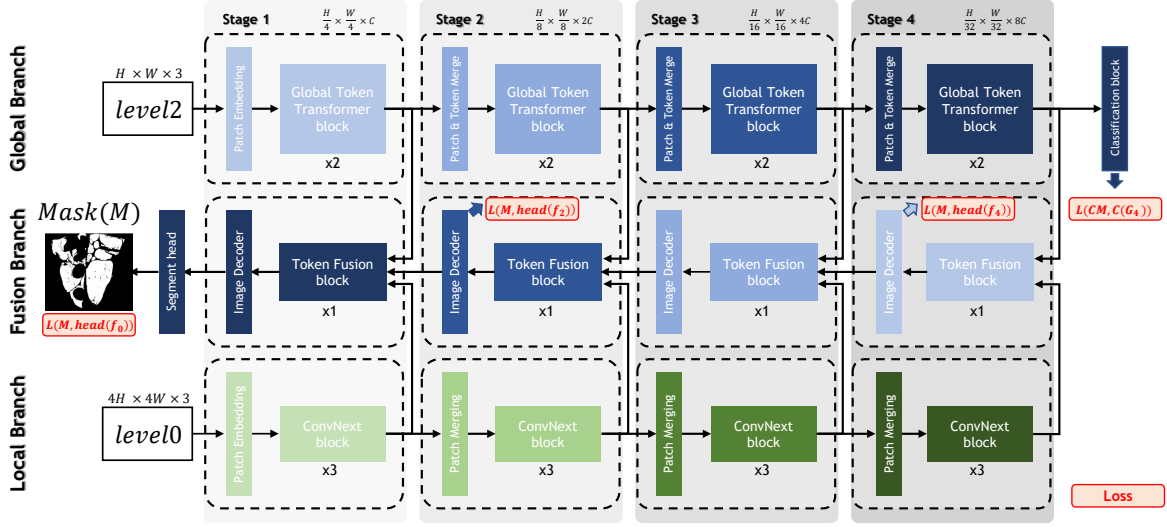


Figure 2. **Overall architecture.** MFTrans operates on two magnification levels (Level 2 and Level 0). The global branch assesses low-magnification features using Global Token Transformer blocks, while the local branch assesses high-magnification features with Local CNN blocks. In the fusion branch, a token fusion block combines features from both branches, and an image decoder generates the final segmentation mask by integrating information from the different magnification levels.

representation extracted from high magnification WSI, such as individual cells, lesions, and cellular morphology, passing through stages of merging layers and ConvNeXt blocks [28]. Lastly, the Token fusion branch integrates the information from the global and local branches, utilizing the features extracted from each stage to provide a comprehensive representation. Through this design, our model is able to effectively integrate global and local representations through a dual-branch architecture, similar to the expert pathology’s diagnosis procedure. In our implementation, each component of MFTrans is comprised of four stages. In the global branch, each stage consists of a patch embedding layer followed by two global token transformer blocks. In the local branch, each stage includes a patch embedding layer followed by three local ConvNeXt blocks. In the fusion branch, a token fusion block and decoder are used at each stage to integrate information from both branches.

3.2. Global branch

3.2.1 Patch Embedding layer

We split the input RGB image (i.e. low magnification levels), $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$, into non-overlapping patches embedding, $\mathbf{w} \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times C}$ using convolution operator. In our implementation, the dimension C is set to 96. After patch embedding, we defined the window as $M \times M$ patches, where M is the size of the window, which yields $N_p = M^2$ patches per window and $N_w = \frac{H}{4 \times M} \times \frac{W}{4 \times M}$ number of windows per image.

3.2.2 Global token-based multi-head self-attention (GT-MSA) Transformer Block

Inspired by GLAM transformer [35], we propose global token-based multi-head self-attention (GT-MSA) to effectively represent global interaction within images (Fig. 3), while significantly reducing the computational cost.

First, we define global tokens $\mathbf{g}^l \in \mathbb{R}^{N_w \times N_g \times C}$ for l -th layer, which means every window have own global token, where (N_g) is the number of global tokens per window, and (C) is the patch embedding dimension. Then, we concatenate the global token with their window, where it yields as $\mathbf{Z}^l \in \mathbb{R}^{N_w \times (N_g + N_p) \times C}$.

Second, we apply the multi-head self-attention (MSA) on $\mathbf{Z}^l$ to compute $\hat{\mathbf{Z}}^l$. Then, GT-MSA is applied to update the global token. GT-MSA uses the MSA for every global token in the window, as illustrated in Fig. 3-(a). The formulation of GT-MSA is described as follows:

$$\text{GT-MSA}(\mathbf{g}^l) = [\hat{g}_1^l, \hat{g}_2^l, \dots, \hat{g}_n^l] \in \mathbb{R}^{N_w \times N_g \times C}, \quad (1)$$

where $\hat{g}_n^l = \text{MSA}(g_n^l)$ ($n = 1, 2, \dots, N_g$), and $g_n^l \in \mathbb{R}^{N_w \times C}$ denotes n -th global tokens for all window. Finally, GT-MSA transformer blocks are computed as

$$\begin{aligned} [\hat{\mathbf{G}}^l; \hat{\mathbf{w}}^l] &= \text{W-MSA}(\text{LN}(\mathbf{Z}^l)), \\ \mathbf{G}^{l+1} &= \text{GT-MSA}(\hat{\mathbf{G}}^l), \\ \mathbf{w}^{l+1} &= \mathbf{w}^l + \text{MLP}(\text{LN}(\hat{\mathbf{w}}^l)), \\ [\mathbf{G}^{l+1}; \mathbf{w}^{l+1}] &= \mathbf{Z}^{l+1}, \end{aligned} \quad (2)$$

where $\mathbf{w}^{l+1} \in \mathbb{R}^{N_w \times N_p \times C}$ denotes global feature and $\mathbf{G}^{l+1}$ denotes concatenated global tokens for all windows (Fig. 3-(a)).

For the computational complexity, the computational complexity of GT-MSA is less than that of GLAM, since GT-MSA computes self-attention for each global token. The computational complexity of GT-MSA is $\Omega_{GT-MSA}(N_g \times (4 \times N_w \times C^2 + 2 \times N_w^2 \times C))$, where $4 \times N_w \times C^2$ represents the number of operations required to generate the Query, Key, and Value matrices, and $2 \times N_w^2 \times C$ denotes the number of operations for calculating the results through the multiplication of attention weights and the Value matrix. For GLAM, the computational complexity is $\Omega_{GLAM}(4 \times N_w \times N_g \times C^2 + 2 \times (N_w \times N_g)^2 \times C)$. The detailed computational complexity of models is presented in the Supplementary S1.

3.2.3 Token & patch merging layer

To model learning a hierarchical representation, patches and global tokens are integrated based on the token & patch merging layer. Briefly, the patch merging layer consisted of convolution operation with 2×2 kernel, stride 2, and the number of filter $2C$. The global token for 2×2 neighboring windows is merged based on channel-wise concatenation and linear projection to produce a $2C$ -dimensional concatenated global token for each channel. The token & patch merging layer follows the GT-MSA transformer to produce hierarchical representation. So, Each time a token passes through the token & patch merging layer, the stage deepens. The first stage, with the patch embedding layer, creates N_w . In the subsequent stages, N_w is reduced by a factor of $1/4$, and C is doubled. So each t -th stage feature denoted by $\mathbf{Z}_t \in \mathbb{R}^{\frac{N_w}{4^{t-1}} \times (N_p + N_g) \times 2^{t-1} C}$. In our implementation, we use the token & patch merging layer instead of the patch embedding layer in all stages except the first.

3.2.4 Classifier block

In the last stage, we add an auxiliary network (i.e., the classifier block) to enhance discrimination in the lower stages of our model [34]. We utilize an attention-based MLP to classify whether each window originates from a tumor or normal tissue. During training, the loss from this auxiliary network is combined with the total loss of the model. However, the classifier block is not used during the inference phase. The detailed description is presented in the Supplementary S2.

3.3. Local branch

3.3.1 Patch Embedding

Similar to Sec. 3.2.1, we split input RGB image (i.e., high magnification levels), where the input image in the local branch being four times larger than in the global branch, $\mathbf{x} \in \mathbb{R}^{4H \times 4W \times 3}$, into non-overlapping patches embedding (i.e., $\mathbf{s} \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times C}$) using convolution operator.

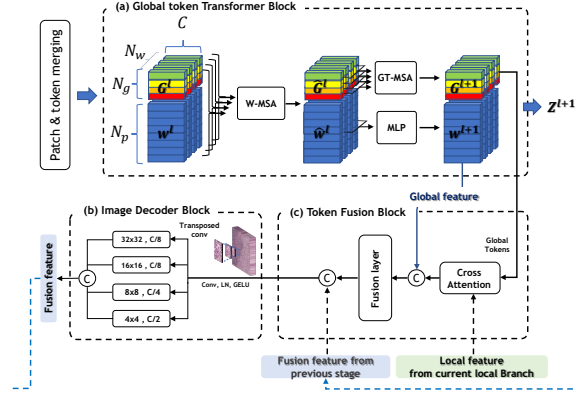


Figure 3. **Detailed architecture of the Global and Fusion branch.** Sub-figures (a), (b), and (c) denote the Global Token Transformer Block, Image Decoder Block, and Token Fusion Block, respectively. This diagram illustrates how the model fuses local and global features from high- and low-magnification images, enabling segmentation in a manner similar to the approach used by pathologists.

3.3.2 ConvNeXt block

We employ ConvNeXt as the backbone of the local image encoder. The ConvNeXt has the advantage of providing CNNs' locality inductive bias, hierarchical feature representation, and parameter efficiency. [28]. In our implementation, the ConvNeXt consists of several layers, including a depth-wise convolution layer(DConv) with a 7×7 kernel size and C filters, Layer Normalization (LN) layer, a 1×1 point-wise convolution(PConv) layer with $4C$ filters, the GELU activation, and PConv layer with C filters, are applied. Finally, a skip connection adds the input of the block to the final output, preserving the spatial information and mitigating the gradient vanishing problem.

$$\begin{aligned} \hat{\mathbf{s}}^l &= \text{LN}(\text{DConv}(\mathbf{s}^l)), \\ \mathbf{s}^{l+1} &= \mathbf{s}^l + \text{PConv}(\text{GELU}(\text{PConv}(\hat{\mathbf{s}}^l))) \end{aligned} \quad (3)$$

where $\mathbf{s}^{l+1}$ denotes local feature, presented in Fig. 3-(c).

3.3.3 Patch merging layer

Similar to the Sec. 3.2.3, we perform convolution operation with 2×2 kernel, stride 2, and the number of filter $2C$ to learn a hierarchical representation. This allows the encoder to capture local features from high magnification images efficiently. Thus, the local feature at t -th stage are :

$$\mathbf{s}_t \in \mathbb{R}^{\frac{H}{2^{t+1}} \times \frac{W}{2^{t+1}} \times 2^{t-1} C} = \mathbb{R}^{\frac{N_w}{4^{t-1}} \times N_p \times 2^{t-1} C} \quad (4)$$

In our implementation, we use the patch merging layer instead of the patch embedding layer in all stages except the first.

3.4. Fusion branch

3.4.1 Token fusion block

In order to integrate multi-resolution WSI, we propose the token fusion block based on a cross-attention mechanism and apply it to each stage, as shown in Fig. 3-(c). Briefly, we first assess globally important local features by using a cross-attention mechanism, where the global token is used as the query, and the local feature serves as the key and value, denoted as “cross attention” in Fig. 3-(c). Next, we concatenate it with the global feature and integrate them based on inverted bottleneck structure [33], which consists of a 5x5 convolution, LN, PConv for channel expansion, a GELU activation function, and another PConv to compress the channels again, denoted as “Fuse block” in Fig. 3-(c). Finally, the output of the Fuse block is concatenated with the result from the previous Image decoder block, described in below.

3.5. Image decoder block

Inspired by DHUNet [38], we implement an image decoder to expand the image features, allowing them to be used in subsequent stages to account for the dimensional differences between each stage, as shown in Fig. 3-(b). For instance, in the first stage, the image decoder transforms the shape from $h_t \times w_t \times C_t$ to $4h_t \times 4w_t \times C_t$. In the following stages, the shape changes from $h_t \times w_t \times C_t$ to $2h_t \times 2w_t \times \frac{1}{2}C_t$. Specifically, the image decoder consists of multiple transposed convolution layers, and the resulting feature maps are concatenated as the output, which is then used as the previous feature of “Fusion stage” in the subsequent stage, as shown in Fig. 3-(c). The details are presented in Supplementary S3.

$$\mathbf{D}_t = \text{Image decoder}[\mathbf{D}_{t+1}; \mathbf{F}_t], \quad (5)$$

where $\mathbf{D}_{t+1}$ denotes the previous fusion stage’s output and $\mathbf{F}_t$ denotes Fuse block’s output. The final output $\mathbf{D}_1$ goes to a 1x1 convolution to make the final segmentation map.

3.6. Loss Function

We train our model in an end-to-end manner using a composite loss function, as presented in Eq. (6). The segmentation loss function, $\mathbf{L}$, is defined as the sum of binary cross-entropy loss ($\mathbf{L}_{bce}$) and Dice loss ($\mathbf{L}_{dice}$), calculated based on the model’s predictions and the ground truth. To further guide the learning process, we incorporate additional auxiliary segmentation losses from intermediate stages, along with a classification loss from the classifier block in the final stage. During training, these auxiliary losses are combined with the total loss, though they are not used during the inference phase. The overall training loss is then defined as:

$$\begin{aligned} \mathbf{L}_{total} = & \alpha \mathbf{L}(M, \text{head}(f_0)) + \beta \mathbf{L}(M, \text{head}(f_2)) \\ & + \gamma \mathbf{L}(M, \text{head}(f_4)) + \omega \mathbf{L}_{bce}(CM, C(G_4)) \end{aligned} \quad (6)$$

where α, β, γ and ω are tunable hyperparameters. The segmentation head generates pixel-wise predictions, and M denotes the ground truth mask. CM is a binary classification mask that indicates 1 or 0 whether the presence or absence of tumor in the ground truth mask.

4. Experiments

In this section, we conduct empirical experiments to provide the promise of integrating multi-resolution based on the dual approach in tumor segmentation. The dataset and experimental settings details are described in Sec. 4.1. In section Sec. 4.2, we present the three experiment results. First, we present the results of three experiments. First, we evaluate the segmentation performance of the proposed method for breast cancer metastases in lymph nodes using the Camelyon 16 dataset. Next, we evaluate generalization of our model across different dataset (i.e., Catholic Uijeongbu St. Mary’s hospital dataset). Finally, we demonstrate robustness of our model using data from entirely different organ (i.e., PAIP dataset). Finally, We conduct ablation studies to validate our method in section Sec. 4.3.

4.1. Experimental setups

4.1.1 Data description

The Camelyon16 dataset The Camelyon16 dataset [3] is part of the Camelyon Grand Challenge 2016, which focused on the detection of metastases in sentinel lymph node whole-slide images from two independent datasets. It includes 400 H&E-stained WSIs (i.e., 270 training images and 130 testing images). The images are annotated with lesion-level ground truth, providing a robust dataset for evaluating algorithms for metastasis detection. The high resolution and detailed annotations of the Camelyon16 dataset make it an ideal benchmark for testing the effectiveness of segmentation algorithms in identifying metastatic regions.

Catholic Uijeongbu St. Mary’s hospital dataset We include a total of 83 WSIs of lymph nodes from 36 patients undergoing either radical mastectomy or breast-conserving surgery with sentinel lymph node biopsy at Catholic Uijeongbu St. Mary’s hospital between January 2023 and December 2023. Of these, 59 WSIs from 22 patients show evidence of metastatic carcinoma, while 24 WSIs from 14 patients exhibit normal lymph node histology. The sizes of the metastatic lesions range from 0.03 mm to 15.4 mm. Specifically, 49 WSIs are classified as macrometastatic (> 2 mm), 8 WSIs as micrometastatic (> 0.2 mm and ≤ 2.0 mm or ≤ 0.2 mm and > 200 cells), and 2 WSIs as containing isolated tumor cells (≤ 0.2 mm and ≤ 200 cells). An expert

pathologist (K.Y) performs the annotations for all cancer regions. This study is approved by the Institutional Review Board of the College of Medicine at the Catholic Uijeongbu St. Mary’s hospital (Approval No. XC23SIDI0021).

The PAIP2019 dataset The PAIP2019 dataset [21] is from the Pathology AI Platform Challenge 2019, which aims to promote the development of algorithms for automated liver cancer segmentation in WSI. This dataset consists of 50 WSIs of liver biopsies stained with H&E, split into training and testing sets. Each image is annotated with detailed segmentation masks for tumor regions, offering a challenging dataset for evaluating segmentation performance. The complexity and variety of tumor appearances in the PAIP2019 dataset provide a rigorous test for the generalizability and robustness of segmentation models.

4.1.2 Implementation details

1) CNN-based methods: We compare our method with CNN-based models, such as Unet **2) Transformer-based methods:** Since the ViT-based model successfully segments objects both natural images and medical images, we compare MFTrans with transformer-based models, including DStansUnet, SwinUnet, and SwinUnetR. **3) Dual branch methods:** Recently dual-branch approaches, including DHUnet, and TransFuse, have shown successful segmentation performance in WSI applications. We compare our method with dual branch models. Note that the major difference between existing dual-branch approaches and our method is that our method utilizes multi-resolution (i.e., 10x and 40x magnification WSI) while existing methods focus on one particular resolution, such as 10x alone.

For each model, we train on 6 Nvidia H100 GPU with 80GB of memory with python 3.10 and pytorch 2.0.0. The hyper-parameters are tuned in certain ranges as follows: learning rate in {0.003, 0.001, 0.0003, 0.0001, 0.00003, 0.00001} and AdamW optimizer with $\alpha = 0.99$, and $\beta = 0.99$, and crossentropy as loss function with $\alpha = 0.6$, $\beta = 0.2$, $\gamma = 0.2$ and $\omega = 0.2$. For training each model,

	Dice	Jaccard	Accuracy
MFTrans(ours)	0.918	0.848	0.925
DHUnet	0.907	0.830	0.915
DStansUnet	0.906	0.829	0.912
TransFuse	0.920	0.851	0.925
SwinUnet	0.848	0.737	0.873
SwinUnetR	0.793	0.657	0.831
Unet	0.857	0.751	0.88

Table 1. Comparison with competing methods on Cameylon 16 for balanced setup (i.e., normal:tumor=1:1). The tumor segmentation performance are compared in terms of dice, Jaccard index, and accuracy. Top results are reported in **bold**.

we use data augmentations including horizontal flip, vertical flip, random [90°, 180°, 270°] rotation, and random hue-saturation-value (HSV) transform with hue 30%, saturation 10%. When we training Transfuse and DStansUnet, the input size of the two models is resized to the size used in the paper respectively.

4.2. Experimental results and discussion

4.2.1 Evaluation on Camelyon 16 dataset

To demonstrate the potential of our method for tumor segmentation, we conduct a quantitative comparison with balanced data setup (i.e., normal:tumor=1:1). As shown in Tab. 1 and Fig. 4, we observe that TransFuse shows the best segmentation performance (i.e., dice= 0.920, Jaccard index= 0.851, accuracy: 0.925), and our method achieves second best performance (i.e., dice= 0.918, Jaccard index= 0.848, accuracy: 0.925) compared with all state-of-the-art (SOTA) methods. Overall, the transformer-based segmentation methods, including DHUnet, DStansUnet, TransFuse and ours, shows better performance than CNN-based segmentation model, such as U-Net, and Swin transformer based model shows the worst performance.

However, our method (i.e., MFTrans) shows promising results for tumor segmentation in real world setup. Particularly, in camelyon 16 dataset, WSI scans contain more normal patches than tumor patches (i.e, normal:tumor=5:1). As shown in Tab. 2 We observe that our method achieves the best performance (i.e., dice= 0.818, Jaccard index= 0.692, accuracy: 0.938) and Transfuse achieves second best performance (i.e., dice= 0.810, Jaccard index= 0.681, accuracy: 0.933) compared with all SOTA methods.

4.2.2 Evaluation on Catholic Uijeongbu St. Mary’s hospital dataset

To assess the generalizability of our method, we conduct a quantitative comparison in Catholic Uijeongbu St. Mary’s hospital dataset, where WSIs are collected from the same

	Dice	Jaccard	Accuracy
MFTrans(ours)	0.818	0.692	0.938
DHUnet	0.802	0.669	0.932
DStansUnet	0.765	0.619	0.913
TransFuse	0.810	0.681	0.933
SwinUnet	0.780	0.640	0.933
SwinUnetR	0.688	0.525	0.902
Unet	0.801	0.668	0.940

Table 2. Comparison with competing methods on Cameylon 16 in real-world setup (i.e., normal:tumor=5:1). The tumor segmentation performance are compared in terms of dice, Jaccard index, and accuracy. Top results are reported in **bold**.

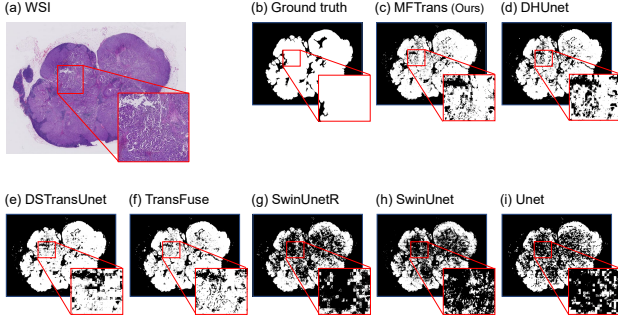


Figure 4. **Visualization of segmentation map for SOTA methods on the Camelyon 16 dataset.** Sub-figures (a) denotes ground truth WSI and its tumor is marked with black solid contour. The segmentation map for benchmark algorithms, including proposed, DHUNet, DStansUnet, TransFuse, SwinUnetR, SwinUnet, Unet are presented in sub-figure (b)-(i), respectively.

organ with Camelyon16. This allows us to evaluate our model on independently collected data under similar conditions. As shown in Tab. 3 and Fig. 5, we observe that our MFTrans shows the best segmentation performance (i.e., dice= 0.902, Jaccard index= 0.829, accuracy: 0.995), and SwinUnet achieves second best performance (i.e., dice= 0.897, Jaccard index= 0.820, accuracy: 0.995) compared with all SOTA methods.

4.2.3 Evaluation on PAIP dataset

To further assess the generalizability across different organs, we conduct a quantitative comparison in the PAIP dataset, where WSIs are collected from the different organs with Camelyon16. Segmenting tumors in WSI from different organs is more challenging due to variations in tissue morphology and staining. Despite this, as shown in Tab. 4 and Fig. 6, we observe that our model shows the best segmentation performance (i.e., dice= 0.801, Jaccard index= 0.688, accuracy: 0.905) compared with all SOTA methods.

	Dice	Jaccard	Accuracy
MFTrans(ours)	0.902	0.829	0.995
DHUnet	0.894	0.817	0.995
DStansUnet	0.860	0.809	0.995
TransFuse	0.893	0.818	0.995
SwinUnet	0.897	0.820	0.995
SwinUnetR	0.850	0.767	0.994
Unet	0.875	0.792	0.994

Table 3. **Comparison with competing methods on Catholic Uijeongbu St. Mary’s hospital dataset.** The tumor segmentation performance are compared in terms of dice, Jaccard index, and accuracy. Top results are reported in **bold**.

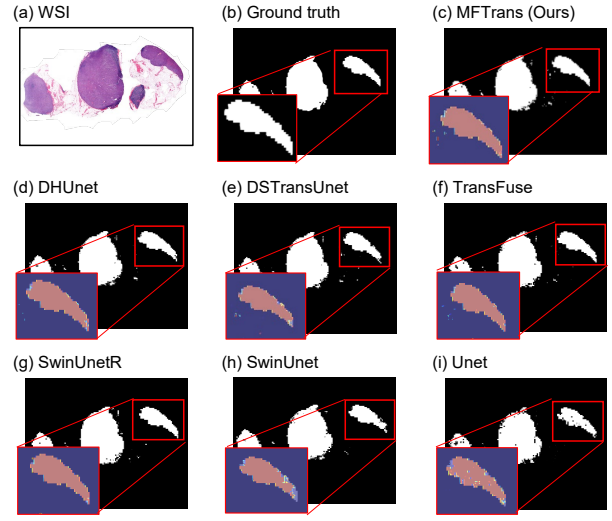


Figure 5. **Visualization of segmentation map for SOTA methods on the Catholic Uijeongbu St. Mary’s hospital dataset.** Sub-figure (a) shows the original WSI, and sub-figure (b) presents the ground truth mask. Segmentation results for benchmark algorithms, including MFTrans (ours), DHUNet, DStansUnet, TransFuse, SwinUnetR, SwinUnet, and Unet, are shown in sub-figures (c) - (i), respectively.

4.2.4 Discussion

While individual performance within each dataset is critical, demonstrating generalization across different datasets is crucial for validating a model’s real-world applicability. The Camelyon 16 and Catholic Uijeongbu St. Mary’s hospital datasets both consist of WSIs from the same organ (i.e. lymph nodes), enabling us to assess the robustness of our model across datasets with similar tissue types but different data sources. Our method perform consistently well in both datasets, indicating that it can generalize effectively even when WSIs are collected independently under similar conditions.

More challenging is generalizing to WSIs collected from entirely different organs, as in the PAIP dataset (e.g., liver).

	Dice	Jaccard	Accuracy
MFTrans(ours)	0.801	0.688	0.905
DHUnet	0.794	0.678	0.904
DStansUnet	0.792	0.673	0.899
TransFuse	0.748	0.610	0.889
SwinUnet	0.773	0.648	0.896
SwinUnetR	0.741	0.608	0.893
Unet	0.767	0.648	0.921

Table 4. **Comparison with competing methods on PAIP dataset.** The tumor segmentation performance are compared in terms of dice, Jaccard index, and accuracy. Top results are reported in **bold**.

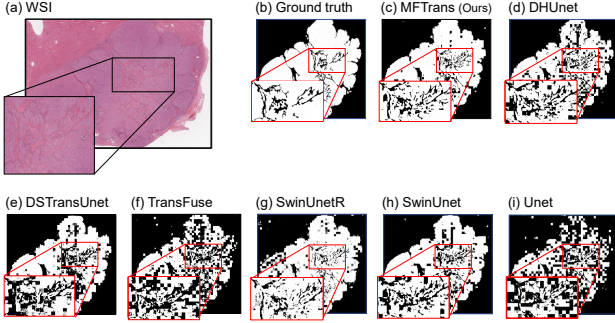


Figure 6. **Visualization of segmentation maps for state-of-the-art (SOTA) methods on the PAIP dataset.** Sub-figure (a) shows the original WSI, and sub-figure (b) presents the ground truth mask. Segmentation results for benchmark algorithms, including MFTrans (ours), DHUNet, DSTransUnet, TransFuse, SwinUnetR, SwinUnet, and Unet, are shown in sub-figures (c) - (i), respectively.

Although differences in tissue morphology typically lead to significant drops in segmentation performance, our method achieves the best performance compared with competing methods in the PAIP dataset. This impressive performance demonstrates its ability to adapt to different tissue types and maintain strong segmentation capabilities. These results suggest that MFTrans is effective within specific datasets and highly generalizable across various WSI data from different organs.

4.3. Ablation Studies

Effect of sample imbalance varying proportion of tumor We evaluate the effect of varying tumor-to-normal tissue ratios on segmentation performance. As shown in Fig. 7, our method outperforms state-of-the-art models, particularly when the dataset contains more than twice as many normal patches as tumor patches. This demonstrates the advantage of our approach in real-world scenarios (i.e., normal:tumor=5:1), where datasets typically have a higher proportion of normal tissue compared to tumor regions.

Effect of Global token number The number of global tokens plays a crucial role in the model’s ability to capture integrated context within the global branch and in selecting important features from the local branch during fusion. As shown in Tab. 5, using too many tokens can degrade performance, while 9 tokens provide the best results. Fewer tokens struggle to capture sufficient global context, whereas increasing model complexity and leading to overfitting on less relevant features. Additional experimental results are provided in the Supplementary S4.

5. Conclusion

In this study, we propose MFTrans, a multi-resolution fusion transformer designed to combine high- and low-

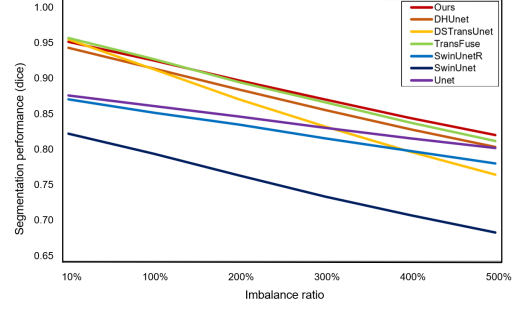


Figure 7. **The effect of tumor-to-normal tissue ratio.** We evaluate segmentation performance with varying imbalance ratios, in terms of Dice coefficients. The imbalance ratio is defined as the number of normal patches divided by the number of tumor patches.

	7	8	9	10	11	12
Dice	0.907	0.894	0.917	0.910	0.909	0.898
Jaccard	0.829	0.808	0.848	0.836	0.832	0.815

Table 5. **Effect of number of Global token.** We compare segmentation performance varying number of Global token in Camelyon16

magnification WSI for better tumor segmentation. The combination of a Global Token Transformer and a dual-branch architecture allows MFTrans to capture both local and global features efficiently, mimicking the hierarchical approach of pathologists. Our extensive experiments on the Camelyon16, PAIP2019, and Catholic Uijeongbu St. Mary’s hospital datasets show that MFTrans outperforms existing state-of-the-art models in terms of segmentation accuracy, especially in real-world cases where tumor-to-normal tissue ratios are imbalanced. The model’s ability to perform well across different datasets and tissue types also highlights its promise for broader use in clinical cancer diagnostics.

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